DRUGS USED IN INFERTILITY

LUKE LUNDAU BIETE

DipPharm, BPharm, MclinPharm

OVERVIEW

- Infertility is a condition of the reproductive system defined by the failure to achieve a clinical pregnancy

- This is after 12 months or more of regular unprotected sexual intercourse

Types of infertility

- There are two kinds of infertility namely;

i. Primary

ii. Secondary

Types of infertility

Primary infertility

- This involves the couple who have never conceived

Secondary infertility

- This involves the couple that has experienced a pregnancy before but failed to conceive later

- The inability to have children, whether primary or secondary affects couples and causes emotional and psychological distress in both men and women

- Globally, most infertile couples suffer from primary infertility

Symptoms of infertility

- Inability to become pregnant
- Inability to maintain a pregnancy
- Inability to carry a pregnancy to a live birth

Causes of infertility

- Infertility can be caused by both men and women factors, with individual sexes accounting for a third of infertility problems respectively

- In remaining cases infertility may be due to problems in both partners or the cause is unclear

Female infertility can be caused by a number of factors

- **Damage to fallopian tubes -** caused by Pelvic inflammatory diseases (PID) caused by various infections, endometriosis, pelvic surgery

- **Disturbed ovarian function/hormonal causes:** Synchronized hormonal changes occur during the menstrual cycle leading to the release of an egg from the ovary (ovulation) and the thickening of the endometrium (inner lining of the uterus) in preparation for the fertilized egg (embryo) to implant inside

  - Any disturbance like noted below will affect fertility;

a) **Polycystic ovary syndrome(PCOS) -** interferes with normal ovulation and is known to be the common cause of female infertility

b) **Functional hypothalamic amenorrhea -** Excessive physical (common in athletes) or emotional stress may result in amenorrhea (absence of periods)

Causes of infertility in females Cont'd

c) Diminished ovarian reserve or premature ovarian aging - women with diminished ovarian reserve may experience difficulty in conceiving

d) Premature ovarian insufficiency - Female ovaries stop working before the age of 40 whose cause can be natural or it can be a disease, surgery, chemotherapy or radiation

- **Uterine causes:** Abnormal anatomy of the uterus; the presence of polyps and fibroids may lead to infertility

- **Cervical causes**: A small group of women may have a cervical condition in which the sperm cannot pass through the cervical canal due to abnormal mucus production or a prior cervical surgical procedure

Male factors causing infertility

- More than 90% of male infertility cases are due to low sperm counts, poor sperm quality or both
- The remaining cases of male infertility can be caused by a number of factors including anatomical problems, hormonal imbalances and genetic defects

Sperm abnormalities include:

- **Oligospermia (low sperm counts) /Azoospermia (no sperms) -** sperm count less than 20 million/ml is termed as oligospermia whereas azoospermia refers to the complete absence of sperm cells in the ejaculate
- **Asthenospermia (Poor sperm motility) -** If 60% or more sperms have abnormal motility (movement is slow and not in straight line) it is termed as asthenospermia and may cause infertility
- **Teratospermia (abnormal sperm morphology) -** about 60% of the sperms should be normal in size and shape for adequate fertility
- Different factors including congenital birth defects, diseases (such as mumps), chemical exposure and life style habits can cause sperm abnormalities.

Treatment of infertility

- Proper history and investigation should be done for effective management
- Supportive treatment in terms of counselling
- Depending on a situation, counselling can be done to a couple or independent parties so to improve on confidentiality

ANTI-ESTROGENS

There are two different subcategories under antiestrogens namely;

- Estrogen receptor antagonists
- Estrogen synthesis inhibitors (Aromatase inhibitors)

Estrogen antagonists

Examples – clomiphene, tamoxifen, raloxifene

- These drugs contain the structure elements required for estrogen receptor binding

- Estrogen antagonists are nonsteroidal drugs that bind to the estrogen and exert tissue specific antagonist and partial agonist effects

1. Clomiphene

Mechanism of action clomiphene

- Clomiphene is a weak estrogen and a moderate estrogen antagonist
- Blocks estrogen receptors in the hypothalamus and the pituitary thereby, preventing the estrogen inhibitory feedback inhibition of gonadotropin secretion
- This leads to subsequent increase in FSH and LH and the subsequent induction of ovarian follicle development and ovulation

Indications

- Anovulatory infertility
- Infertility associated with polycystic ovary disease
- Clomiphene treatment is usually started on the fifth day of the cycle after the start of uterine bleeding and continues for 5 days
- This timing of clomiphene dosing is backed by the fact that ovulation is expected to take place 5 to 10 days after the last dose of clomiphene

NB:

- Ovulation can be detected by checking basal body temperature, urinary LH secretion, plasma progesterone or endometrial history

Side effects

- Multiple births – twins, triplets and so on have been reported to be common
- The risk of multiple births is reduced by starting with lower doses of clomiphene and then increase in each cycle until ovulation occurs

2. Tamoxifen

Mechanism of action

- Has both estrogenic and anti-estrogenic properties
- In the breast tissue, it acts as estrogen receptor antagonist

Indications

- Treatment or prevention of breast cancer in patients with tumors cell that are estrogen receptor positive
- Its use is typically adjuvant in combination with surgery or other chemotherapy
- Tamoxifen inhibits the proliferation of breast cancer cells and may lead to eradication of micrometases
- In advanced or metastatic breast cancer it can used alone as palliative treatment

Side effects

- Nausea and vomiting
- Hot flashes
- Vaginal bleeding
- Menstrual irregularities
- May stimulate proliferation of endometrial cells
- May increase the risk of endometrial cancer

Raloxifene

- A known estrogen receptor modulator

Mechanism of action

- It produces estrogen like effects on the bone and lipid metabolism while antagonizing the effects of estrogen on the breast tissue
- Unlike tamoxifen, raloxifene also acts as an estrogen antagonist in the uterine tissue

Indications

- Prevention of osteoporosis

Caution

- Raloxifene increases the risk of stroke, pulmonary and DVT

Aromatase inhibitors

Examples of aromatase inhibitors – Anastrozole, Letrozole

- Breast cancer is often hormonally associated (hormonally responsive)

- It is therefore referred to as either estrogen or progesterone receptor positive

- Decreasing estrogen levels has been recorded as a very significant intervention in the management of breast cancer

- This can be achieved by either using estrogen antagonists or estrogen synthesis inhibitors (Aromatase inhibitors)

Mechanism of action

- In post menopausal women, estrogen is derived from adrenal androgens, testosterone and androstenediol being primary
- These are then converted to estrogen in the peripheral tissue or cancer tissue itself by enzyme aromatase
- Inhibiting enzyme aromatase and subsequent reduced levels of estrogen is the mechanism by which aromatase inhibitors work

Indications

- First line treatment for locally advanced or metastastic breast cancer in post menopausal women
- Estrogen sensitive breast cancer (Halts progression by lowering estrogen levels)
- Letrozole has recently been used for infertility in anovulatory women either alone or in combination with gonadotropins (better pregnancy rates than clomiphene)

Side effects

Aromatase inhibitors inhibitors are well tolerated but can cause the following;

- Hot flashes
- Nausea
- Headache
- Vaginal bleeding
- Backache

Anti-progestins

- Example is mifepristone

Mechanism of action

- A steroid that acts as progesterone receptor antagonist when progesterone is present or partial agonist in the absence of progesterone
- Also acts as antagonist at the glucocorticoid receptor
- With its action as an antagonist, it causes the breakdown of the endometrium of the pregnant uterus (Decidua) and the subsequent detachment of the blastocycst from the endometrium

Indications

- Medical termination of pregnancy of 49 days of gestation
- For this indication, mifepristone is administered together with misoprostol, a prostaglandin analogue (Different combo brands on the market)
- Mifeprestone is first administerd alone as a single dose which then is followed by misoprostol 48hour after to help stimulate contractions

Side effects of mifepristone

- Anorexia
- Nausea
- Vomiting
- Abdominal pain
- Fatigue
- Heavy uterine bleeding

Androgens

- **Examples** are Testosterone and methyltesteosterone

- Different formulations do exist such as gels, patches for transdermal administration and long acting intramuscular formulations like testosterone cypionate

Indications

- Hypogonadism caused by primary testicular failure or occurring secondary to pituitary insufficiency

- In hypopituitarism, it should be combined with growth hormone to maximize on skeletal growth

- Prevention of osteoporosis in older men

- HRT in postmenopausal women who experience bleeding when only estrogen is used

Danazol

- A synthetic steroid that has anti estrogenic activity

Mechanism of action

- Sends feedback inhibition of gonadotropin secretion and decreased secretion of estrogen

Indications

- Heavy menstrual bleeding
- Endometriosis

Anti-androgens

Gonadotropin releasing Hormone(GnRH) Analogues

- **Examples** – Leuprolide, triptorelin

Mechanism of action

- These drugs act through a variety of mechanisms including inhibition LH secretion, DHT synthesis and antagonism of androgen receptors

Indications

- Inoperable prostate cancer
- Because these drugs can increase LH and testosterone when first administered, they are usually combined with androgen receptor blockers like flutamide, bicalutamide
- This prolong the progression free period and the length of survival in men with advanced prostate

Androgen receptor antagonists

Examples – Bicalutamide, flutamide, nilutamide

Mechanism of action

- These compete with testosterone for the androgen receptor and thus reducing the activity of testosterone

Indication

- Inoperable prostate cancer
- Usually administered with GnRH
- Cyproterone is an androgen that is used for treatment of hirsutism in women and reduce sex drive in men

Side effects

- Nausea
- Gynacomastia
- Impotence
- Hot flashes
- Hepatitis

5α Reductase inhibitors

- Examples – Finasteride and dutasteride

Mechanism of action

- These are synthetic testosterone derivatives that block 5α Reductase the action which reduces the synthesis of DHT in the prostate gland, skin and other target tissues

Indications

- Symptomatic BPH where they improve urinary flow and decrease the risk of urinary retention (Sometimes used together with α adrenoceptor antagonists like tamsulosin)
- Male pattern baldness

Adverse effects

- Erectile dysfunction
- Decreased libido
- Gynecomasticia

END